
Adders

Adders?

Binary Adders are arithmetic circuits in the form of half-adders and full-adders used to add together two binary digits.

Another –

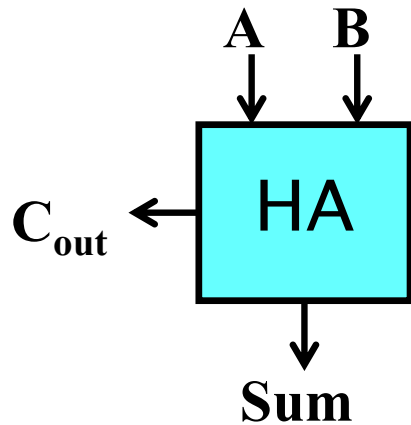
Adder is a combinational logic circuit which can be constructed using just a few basic logic gates allowing it to add together two or more binary numbers is the Binary Adder

Outline

- One-bit adder,
- basic ripple-carry adder
- Carry-Lookahead adders (CLA)
- Brent-Kung adder

Adding Two One-bit Operands

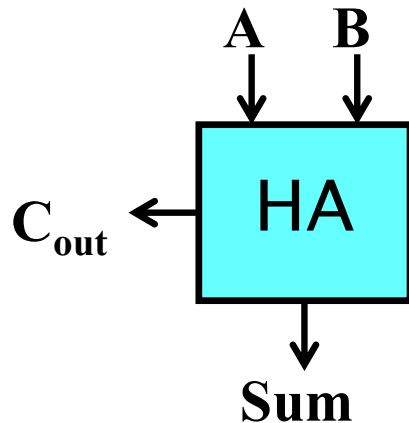
- One-bit Half Adder:



A	B	Sum	Cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Adding Two One-bit Operands

- One-bit Half Adder:



A	B	Sum	Cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

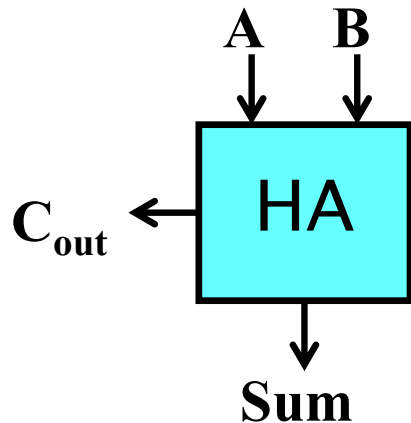
From the truth table

$$\text{Sum} = A'B + AB' = A \oplus B$$

$$\text{Cout} = A.B$$

Adding Two One-bit Operands

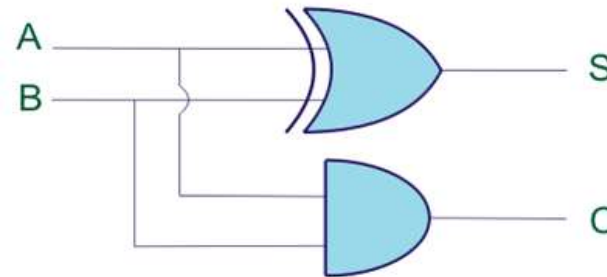
- One-bit Half Adder:



A	B	Sum	Cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

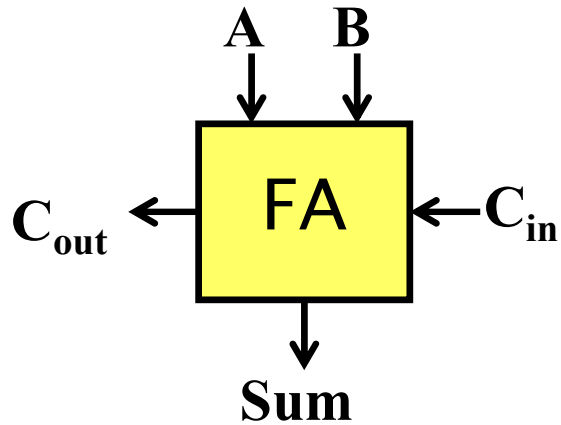
$$\text{Sum} = A'B + AB' = A \oplus B$$

$$\text{Cout} = A.B$$



Adding Two One-bit Operands

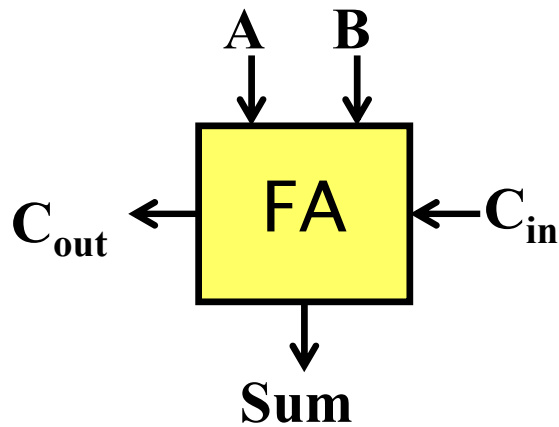
- One-bit Full Adder:



C _{in}	A	B	Sum	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Adding Two One-bit Operands

- One-bit Full Adder:



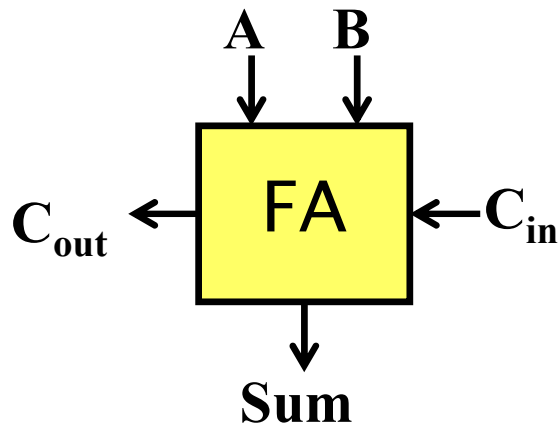
C_{in}	A	B	Sum	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$\begin{aligned}\text{Sum} &= A'BC_{in}' + AB'C_{in}' + A'B'C_{in} + ABC_{in} \\ &= A \oplus B \oplus C_{in}\end{aligned}$$

$$\begin{aligned}C_{out} &= ABC_{in}' + A'BC_{in} + AB'C_{in} + ABC_{in} \\ &= A.B + B.C_{in} + A.C_{in}\end{aligned}$$

Adding Two One-bit Operands

- One-bit Full Adder:

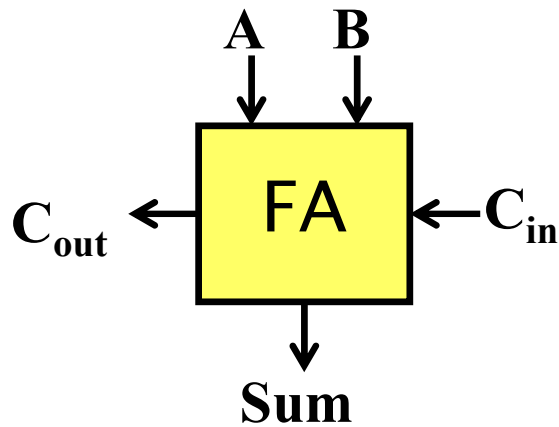


$$\begin{aligned}\text{Sum} &= A'B\text{Cin}' + AB'\text{Cin}' + A'B'\text{Cin} + ABC\text{in} \\ &= A \oplus B \oplus \text{Cin}\end{aligned}$$

C _{in}	A	B	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Adding Two One-bit Operands

- One-bit Full Adder:



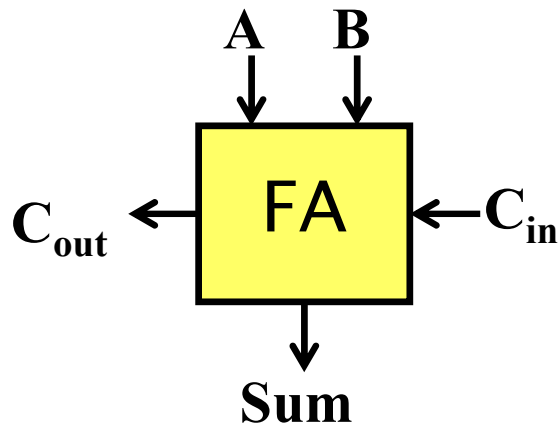
C_{in}	A	B	Sum	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$\begin{aligned}\text{Sum} &= A'BC_{in}' + AB'C_{in}' + A'B'C_{in} + ABC_{in} \\ &= A \oplus B \oplus C_{in}\end{aligned}$$

$$\begin{aligned}S &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + A\overline{B}\overline{C} + A\overline{B}C \\ &= C(\overline{A}\overline{B} + A\overline{B}) + \overline{C}(\overline{A}\overline{B} + A\overline{B}) \\ &= C(\overline{A\overline{B}} + A\overline{B}) + \overline{C}(\overline{A\overline{B}} + A\overline{B}) \\ &= C(\overline{A \oplus B}) + \overline{C}(A \oplus B) = A \oplus B \oplus C.\end{aligned}$$

Adding Two One-bit Operands

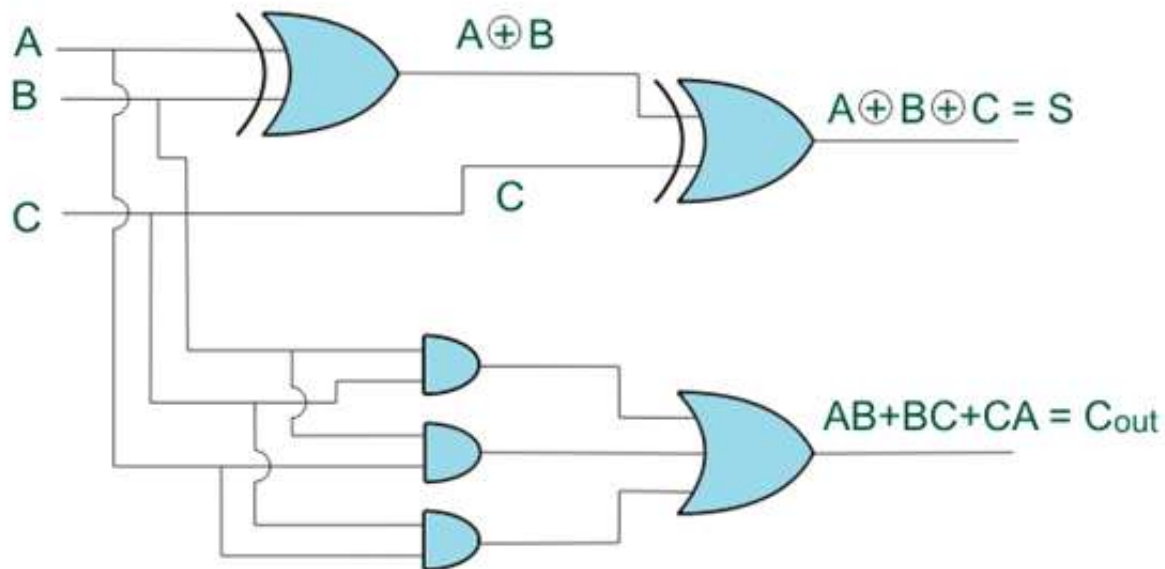
- One-bit Full Adder:



$$\begin{aligned}\text{Sum} &= A'BCin' + AB'cin' + A'B'cin + ABCin \\ &= A \oplus B \oplus Cin\end{aligned}$$

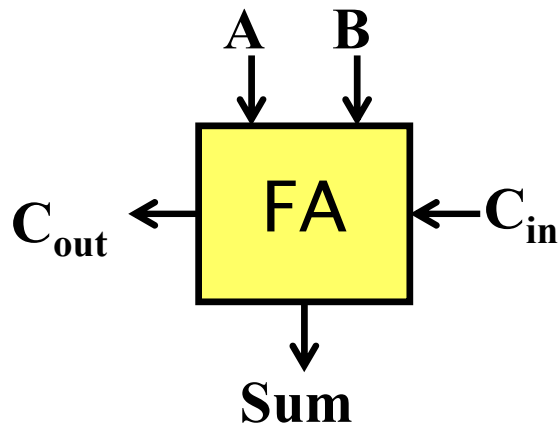
$$\begin{aligned}\text{Cout} &= ABCin' + A'BCin + AB'cin + ABCin \\ &= A.B + B.Cin + A.Cin\end{aligned}$$

C_{in}	A	B	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Adding Two One-bit Operands

- One-bit Full Adder Circuit Using Half-Adder



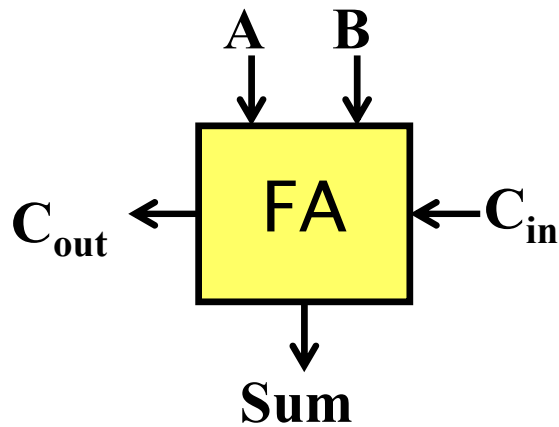
$$\begin{aligned} C_{out} &= ABC_{in}' + A'BC_{in} + AB'C_{in} + ABC_{in} \\ &= A.B + B.C_{in} + A.C_{in} \end{aligned}$$

C _{in}	A	B	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$\begin{aligned} C_{out} &= \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC \\ &= (\overline{A}B + A\overline{B})C + AB(\overline{C} + C) \\ &= (A \oplus B).C + AB. \end{aligned}$$

Adding Two One-bit Operands

- One-bit Full Adder Circuit Using Half-Adder



C_{in}	A	B	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$\begin{aligned}
 C_{out} &= \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC \\
 &= (\overline{A}B + A\overline{B})C + AB(\overline{C} + C) \\
 &= (A \oplus B).C + AB.
 \end{aligned}$$

$$\begin{aligned}
 S &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + A\overline{B}\overline{C} + A\overline{B}C \\
 &= C(\overline{A}\overline{B} + A\overline{B}) + \overline{C}(\overline{A}\overline{B} + A\overline{B}) \\
 &= C(\overline{A} + A)\overline{B} + \overline{C}(\overline{A} + A)\overline{B} \\
 &= C(\overline{A} + A)\overline{B} + \overline{C}(A \oplus B) = A \oplus B \oplus C.
 \end{aligned}$$

Adding Two One-bit Operands

- One-bit Full Adder Circuit Using Half-Adder

$$\begin{aligned}
 S &= \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC \\
 &= C(\overline{A}\overline{B} + AB) + \overline{C}(\overline{A}B + A\overline{B}) \\
 &= C(\overline{A\oplus B}) + \overline{C}(A\oplus B) \\
 &= C(\overline{A\oplus B}) + \overline{C}(A\oplus B) = A\oplus B\oplus C.
 \end{aligned}$$

$$\begin{aligned}
 C_{out} &= \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC \\
 &= (\overline{A}B + AB)C + AB(\overline{C} + C) \\
 &= (A\oplus B).C + AB.
 \end{aligned}$$

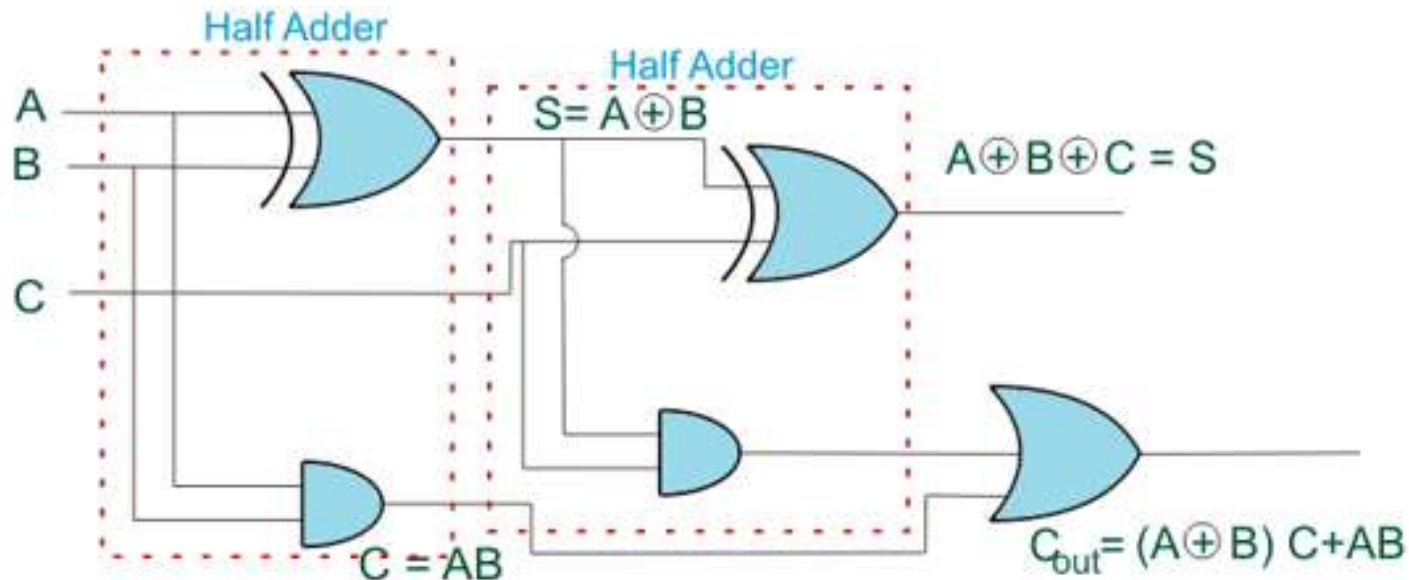
C_{in}	A	B	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Adding Two One-bit Operands

- One-bit Full Adder Circuit Using Half-Adder

$$\begin{aligned} S &= \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC \\ &= C(\overline{A}\overline{B} + \overline{A}B) + \overline{C}(\overline{A}\overline{B} + AB) \\ &= C(\overline{A}(\overline{B} + B) + \overline{C}(\overline{A} + A)) \\ &= C(\overline{A} + \overline{C}) + \overline{C}(A + B) = A \oplus B \oplus C. \end{aligned}$$

$$\begin{aligned} C_{out} &= \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC \\ &= (\overline{A}B + A\overline{B})C + AB(\overline{C} + C) \\ &= (A \oplus B).C + AB. \end{aligned}$$



Thank You